# Touchless remote controller

**Goal:**

- Develop a touchless controller – Reducing the risk of infection during the Covid-19 epidemic.
- Learn about how sensors work and signal processing.

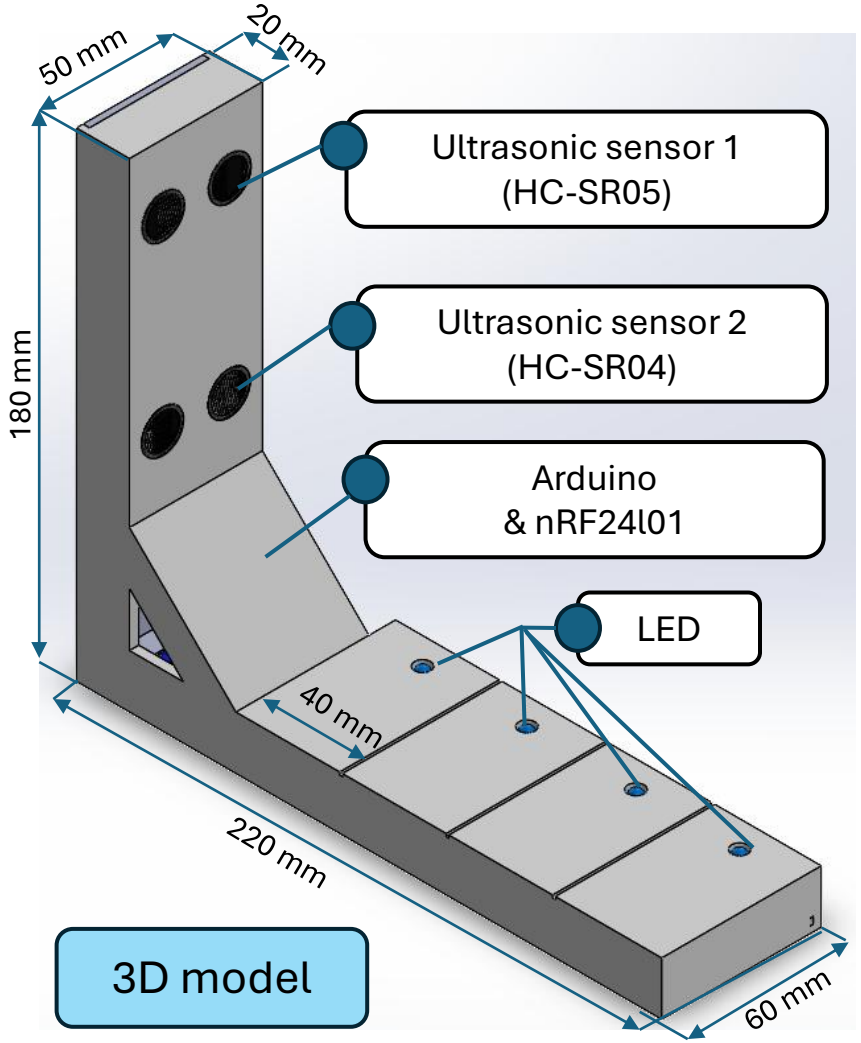
**Idea:**

- Swipe hand across certain distances in front of the sensor.
- With different distances, the controller will send different signals using 2.4KHz band.
- The brightness of the light will show the status of the thing we are controlling

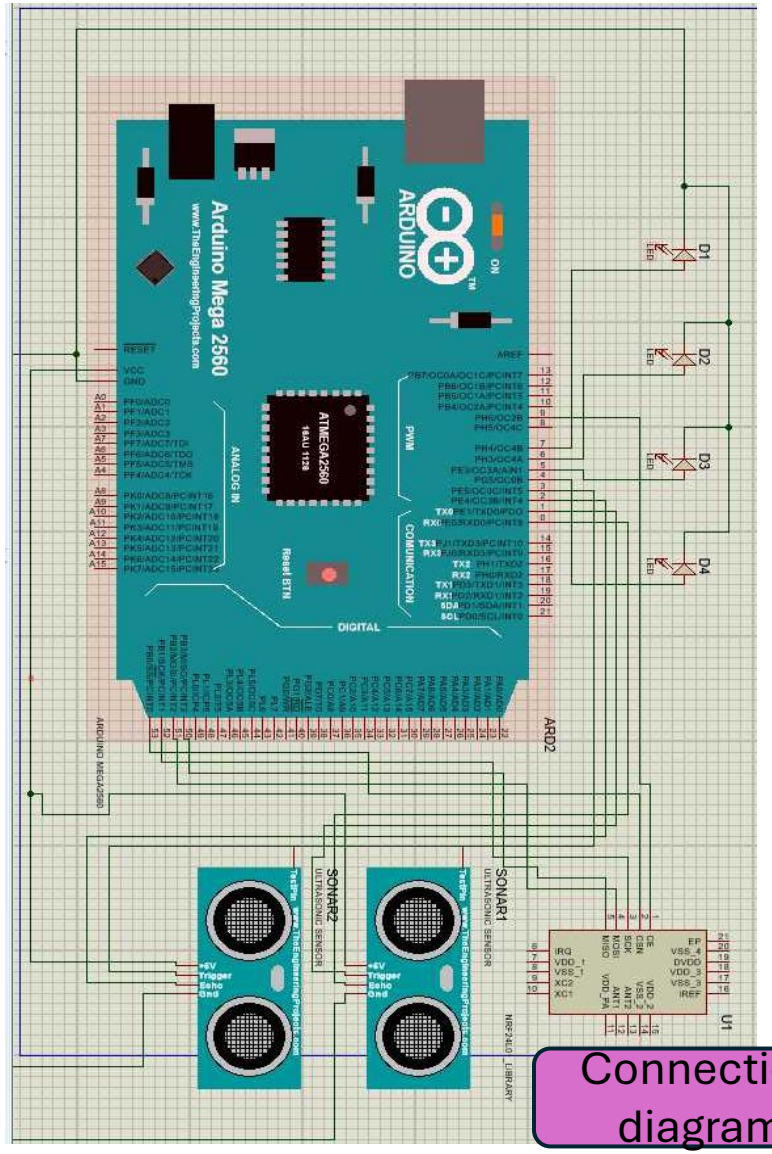


# Touchless remote controller

## ❑ Design




3D model



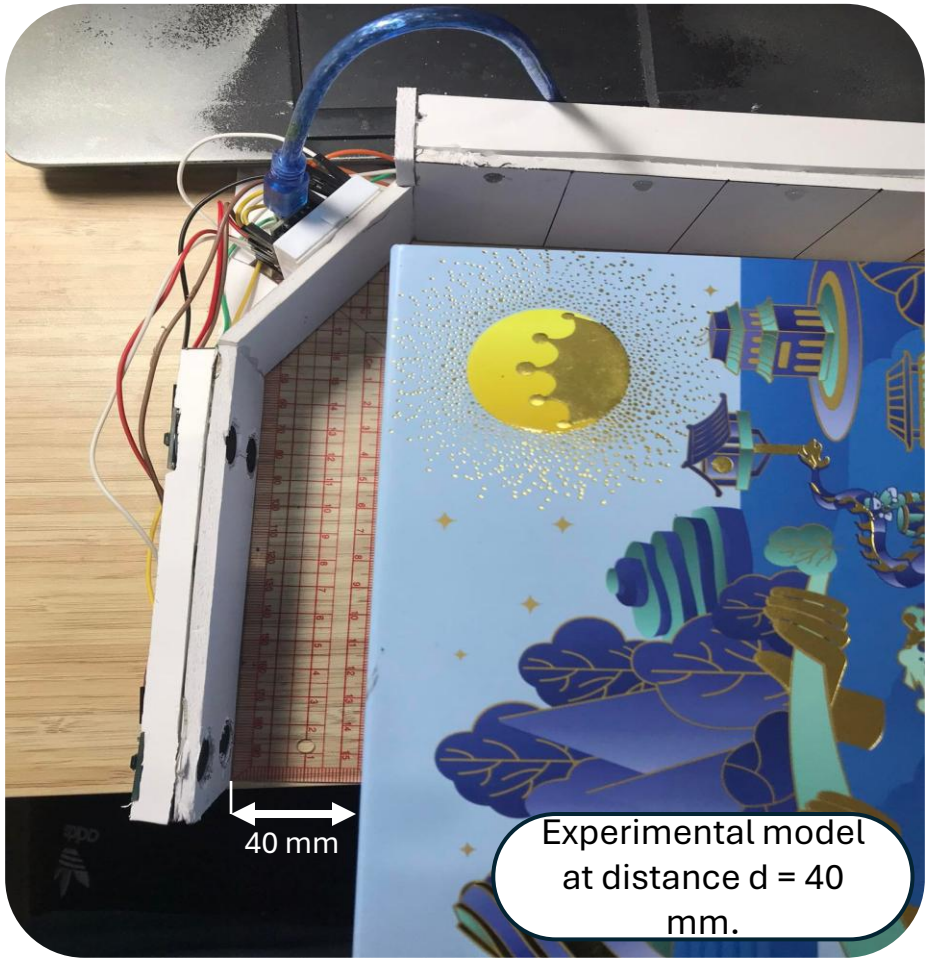

Connection diagram

# Touchless remote controller

## ❑ Experiment and improving accuracy

**Implementation steps:**

- Measure pulse time for different distances (4cm, 8cm, 12cm, 16cm, 20cm) - Each one measure 50 times
- Statistic data and calculate the errors.
- Observe the oscilloscope graph of the sensors.
- Apply Kalman filter
- Compare the results.

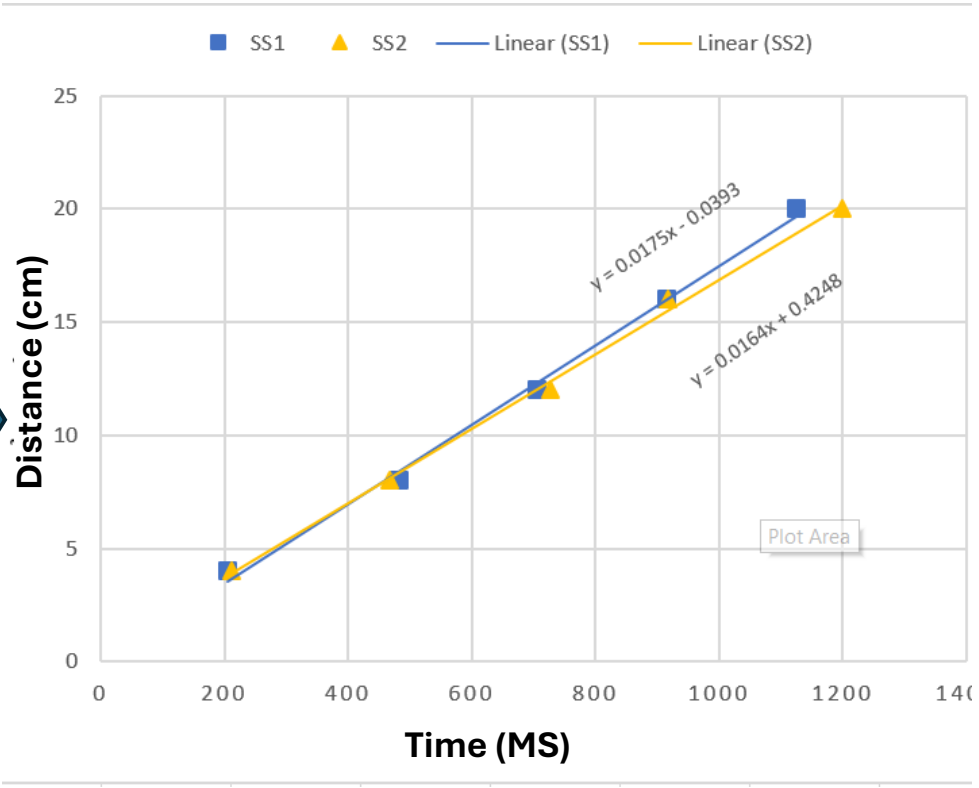



Experimental model at distance d = 40 mm.

# Touchless remote controller

## ❑ Illustrate measured data

| Distance | Time pulse | Frequency | Average | Variance | Standard deviation | Error |
|---|---|---|---|---|---|---|
| 4 | 199 | 4 | 204.84 | 1.8693 | 0.4546 | 11.37% |
| | 205 | 38 | | | | |
| | 207 | 8 | | | | |
| | 206 | 8 | 210.96 | 2.1996 | *0.218552* | 5.46% |
| | 210 | 2 | | | | |
| | 212 | 40 | | | | |
| 8 | 464 | 11 | 483.12 | 13.4516 | 0.4153 | 5.19% |
| | 471 | 9 | | | | |
| | 491 | 17 | | | | |
| | 497 | 8 | | | | |
| | 498 | 5 | | | | |
| | 464 | 12 | 466.84 | 2.9214 | *0.047708* | 0.60% |
| | 465 | 9 | | | | |
| | 466 | 15 | | | | |
| | 471 | 9 | | | | |
| | 472 | 5 | | | | |
| 12 | 684 | 9 | 704.76 | 11.1688 | *0.294* | 2.45% |
| | 690 | 3 | | | | |
| | 708 | 4 | | | | |
| | 709 | 21 | | | | |
| | 714 | 4 | | | | |
| | 715 | 9 | | | | |
| | 725 | 32 | 726.76 | 2.6575 | 0.343664 | 2.86% |
| | 726 | 4 | | | | |
| | 731 | 14 | | | | |
| 16 | 905 | 12 | 914.36 | 12.265 | *0.038* | 0.24% |
| | 906 | 5 | | | | |
| | 907 | 8 | | | | |
| | 908 | 8 | | | | |
| | 912 | 3 | | | | |
| | 932 | 10 | | | | |
| | 938 | 4 | | | | |
| | 917 | 29 | 918.06 | 2.0041 | 0.307178 | 1.92% |
| | 918 | 15 | | | | |
| | 923 | 4 | | | | |
| | 924 | 2 | | | | |
| 20 | 1105 | 8 | 1125.56 | 9.0491 | 0.342 | 1.71% |
| | 1127 | 6 | | | | |
| | 1128 | 5 | | | | |
| | 1130 | 25 | | | | |
| | 1131 | 6 | | | | |
| | 1193 | 1 | 1199.26 | 1.0161 | *0.109262* | 0.55% |
| | 1199 | 30 | | | | |
| | 1200 | 19 | | | | |



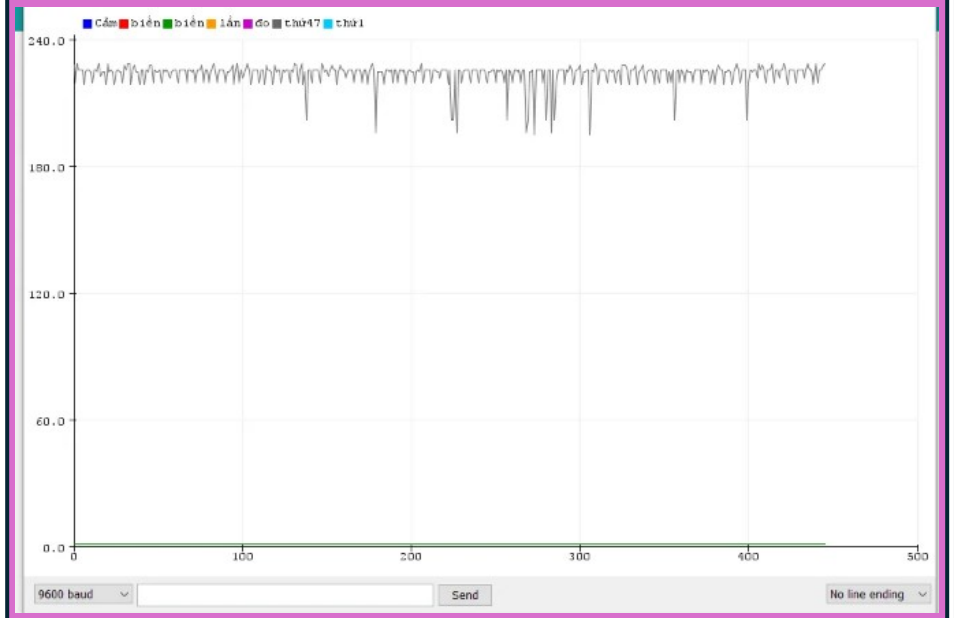

Error chart of 2 sensors

# Touchless remote controller

## ❑ Oscilloscope graph of the sensor

### ❖ Without Kalman filter

Sensor 1




Distance d = 40 mm

Sensor 1



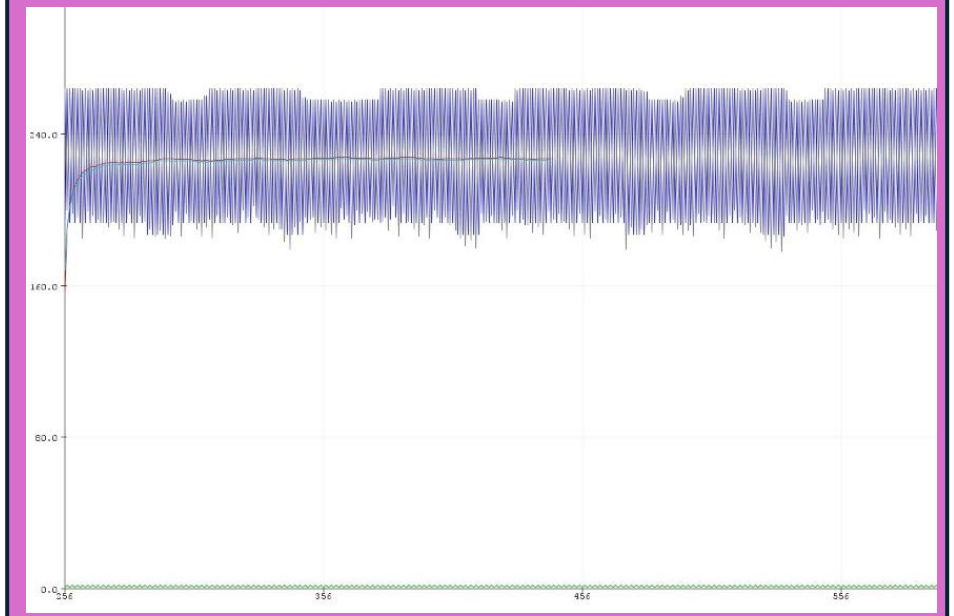

Distance d = 80 mm

# Touchless remote controller

## ❑ Oscilloscope graph of the sensor

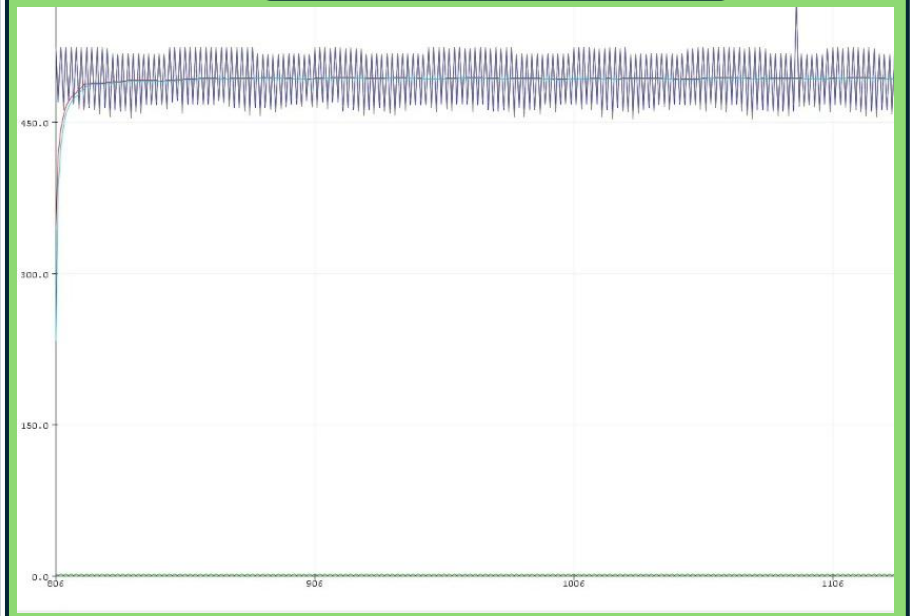
### ❖ With Kalman filter

Sensor 1



Distance d = 40 mm

Sensor 1



Distance d = 80 mm

# Touchless remote controller

❑ Result

**Manufactured controller with two mode:**

**Mode 1:**

**Swipe hand and the corresponding light in that position will light up/darken.**

- Swipe down:
  - ❖ The light gradually brighten
  - ❖ When reach the highest level, if continue to swipe down, the light will flash.

- Swipe up:
  - ❖ The light gradually darkens
  - ❖ When the light is off, if you continue to swipe up, the light will flash



# Touchless remote controller

❑ Result

**Manufactured controller with two mode:**

**Mode 2:**

**Place hand in front of one of the two sensors at different distances to send different signals.**

- Application: Control a dragon-shaped robot
  - ❖ Robot can move, rotate righ-left, lift up- down
  - ❖ Sensor 1 (upper): increase moving speed,turn left, lift down.
  - ❖ Sensor 2 (lower): decrease moving speed, turn right, lift up.

